



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

03-12-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	D	C	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C	B,C	A,B			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	A	B	B	D		
SECTION-II	Q.	1	2	3	4	5	6
	A.	2	3	5	160	12	198

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	B	D	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	B,C,D	C,D	A,B,C,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	A	C	C	A		
SECTION-II	Q.	1	2	3	4	5	6
	A.	9	316	8	15	7	5

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	A	A	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C,D	C,D	A,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	A	A	A	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	518	406	1	9	1	35

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

3. Ans (C)

$$\frac{T}{A} = y \frac{\partial y}{\partial x}$$

$$\frac{\partial y}{\partial x} = 10^{-6} \times \frac{\pi}{2} \cos\left(\frac{\pi}{2} \times \frac{1}{2}\right)$$

$$\frac{T}{A} = 10^{12} \times 10^{-6} \times \frac{\pi}{2\sqrt{2}}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A,B)

$$20 - T_1 = 2a \text{ (for 2 kg block)} \quad \dots (i)$$

$$T_2 - 10 = 1a \text{ (for 1 kg block)} \quad \dots (ii)$$

$$T_1 \frac{1}{2} - T_2 \frac{1}{2} = (2) \left(\frac{1}{2}\right)^2 \frac{1}{2} (2) \frac{a}{\frac{1}{2}} \text{ (for the shaft)}$$

+ disc system)

$$\text{So, } T_1 - T_2 = 2a \quad \dots (iii)$$

After solving these three equation

$$a = 2 \text{ m/s}^2$$

PART-1 : PHYSICS

SECTION-II

3. Ans (5)

$$\Delta S_{\text{ice}} = mC \ln \frac{T_2}{T_1} + \frac{mL_V}{T}$$

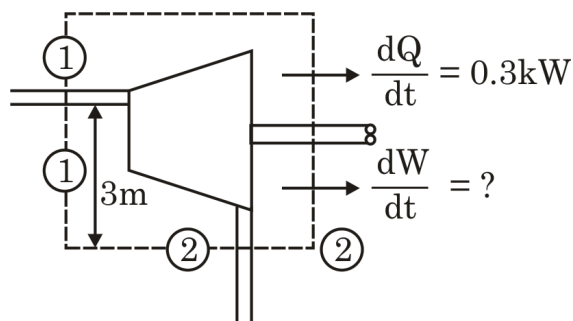
$$= 3 \times 2 \times 10^3 \ln \left(\frac{273}{270}\right) + \frac{3 \times \frac{1000}{3} \times 10^3}{273} + 3$$

$$\times 4.2 \times 10^3 \times \ln \left(\frac{300}{273}\right)$$

$$\approx 4.91 \text{ kJ / K}$$

6. Ans (198)

$$w_1 = w_2 = 0.4 \text{ kg/s}$$



By S.F.E.E.

$$w_1 \left(h_1 + \frac{V_1^2}{2000} + \frac{gZ_1}{1000} \right) + \frac{dQ}{dt}$$

$$= w_2 \left(h_2 + \frac{V_2^2}{2000} + \frac{gZ_2}{1000} \right) + \frac{dW}{dt}$$

$$0.4 \left(3000 \times 10^3 + \frac{900}{2} + 10 \times 3 \right) - 0.3 \times 10^3$$

$$= 0.4 \left(2500 \times 10^3 + \frac{100^2}{2} + 0 \right) + \frac{dW}{dt}$$

$$\frac{dW}{dt} = 0.4 (500 \times 10^3 - 5000 + 480) - 0.3 \times 10^3$$

$$\Rightarrow 197.892 \text{ kJ}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (D)

$$n_{\text{As}} = n_{\text{H}_3\text{AsO}_3} = n \text{ and } n_{\text{eq}} \text{H}_3\text{AsO}_3 = n_{\text{eq}} \text{I}_2 = \frac{Q}{F}$$

$$\frac{w}{75} = \frac{1.68 \times 10^{-3} \times 96.5}{2 \times 96500} \Rightarrow w = 6.3 \times 10^{-5} \text{ gm}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

6. Ans (C,D)

Theoretical.

PART-2 : CHEMISTRY

SECTION-I (iii)

8. Ans (A)

$$(P) K = \frac{K_A}{2} = \frac{0.693}{80} s^{-1}$$

$$r = K[A]_0 = \frac{0.693}{80} \times 0.8$$

$$= 6.93 \times 10^{-3} M/s$$

$$(Q) r_A = \frac{(4-2)}{40 \times 5} = 10^{-2} M/s$$

$$r = \frac{r_A}{2} = 5 \times 10^{-3} M/s$$

$$(R) r = K[A]$$

$$= \frac{0.693}{80} \times 0.04$$

$$= 3.465 \times 10^{-4} M/s$$

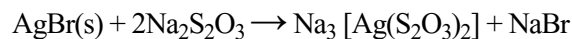
$$(S) KA = \frac{1}{t_{1/2}} \times \frac{1}{[A]_0} = \frac{1}{40} \times \frac{1}{4/5}$$

$$r = \frac{5}{40 \times 4 \times 2} \times (0.008)^2 = 10^{-6} M/s$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (9)



t = 0	0.2 M	0	0
t	0.2 - x	x/2	x/2

$$K_{sp} \times K_f = \frac{\frac{x}{2} \times \frac{x}{2}}{(0.2 - x)^2}$$

$$\text{On solving } x = 0.17716$$

$$\therefore n_{Br^-} = \frac{0.177}{2} = 0.088 \text{ mole/litre}$$

$$\text{in 100 L solution mole of AgBr} = 0.088 \times 100$$

$$= 8.8 \text{ moles}$$

2. Ans (316)

$$k(27^\circ C) = \ln \frac{1}{0.8} = 0.2 \text{ hr}^{-1}$$

$$k(37^\circ C) = \ln \frac{0.8}{0.4} = 0.7 \text{ hr}^{-1}$$

$$\Rightarrow \ln \frac{0.7}{0.2} = \frac{E_a}{R} \left(\frac{10}{300 \times 310} \right) \Rightarrow E_a = 89280 \text{ J}$$

$$k(T) = \ln \frac{0.4}{0.1} = 1.4 \text{ hr}^{-1}$$

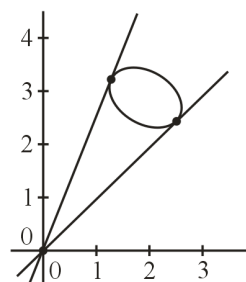
$$\Rightarrow \ln \frac{1.4}{0.7} = \frac{89280}{8} \left(\frac{T - 310}{310 T} \right)$$

$$\Rightarrow \boxed{T = 316.15 \text{ K}}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)



$\frac{y}{x}$ represents the slope of a line passing through the origin.

It follows that since a line $y = mx$ intersects the ellipse at either 0, 1, or 2 points,

the minimum and maximum are given when the line $y = mx$ is a tangent with only one point of intersection. Substituting,

$$2x^2 + x(mx) + 3(mx)^2 - 11x - 20(mx) + 40 = 0$$

Rearranging by the degree of x ,

$$(3m^2 + m + 2)x^2 - (20m + 11)x + 40 = 0$$

Since the line $y = mx$ is tangent to the ellipse, we want the discriminant,

$$(20m + 11)^2 - 4 \cdot 40 \cdot (3m^2 + m + 2)$$

$$= -80m^2 + 280m - 199$$

to be equal to 0. We want $a + b$, which is the sum of the roots of the above quadratic. By Vieta's

$$\text{formulas, that is } \frac{280}{80} = \frac{7}{2}$$

2. Ans (A)

Note that the first five letters must be B's or C's, the next five letters must be C's or A's and the last five letters must be A's or B's. If there are k B's

In the first five letters, then there must be $5 - k$ C's in the first five letters, so there must be k C's and $5 - k$ A's in the next five letters and k A's and $5 - k$ B's in the last five letters. Therefore the number of each letter in each group of five is determined completely by the number of B's in the first 5 letters and the number of ways to arrange these 15 letters

$$\text{with this restriction is } \binom{5}{k}^3 \text{ (since there are } \binom{5}{k} \text{$$

ways to arrange k B's and $5 - k$ C's.

$$\text{Therefore the answer is } \sum_{k=0}^5 \binom{5}{k}^3.$$

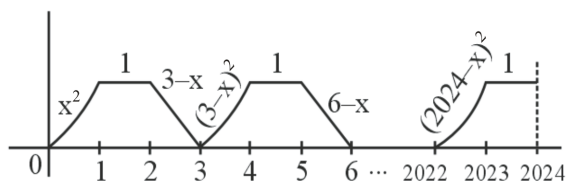
3. Ans (A)

$$g(x) = \begin{cases} 1, & x = 0 \\ 0, & x \neq 0 \end{cases}$$

$$f(x) = \begin{cases} 0, & x \in [0, 1) \\ 1, & x \in [1, 2) \\ 2, & x \in [2, 3) \end{cases}$$

$$\phi(x) = \begin{cases} x^2, & x \in [0, 1) \\ 1, & x \in [1, 2) \\ 3-x, & x \in [2, 3) \end{cases}$$

$$\therefore \int_0^{2012} \phi(x) dx =$$



$$675 \left(\frac{1}{3} \right) + 675 \times 1 + 670 \times \frac{1}{2}$$

$$= 225 + 675 + 335 = 1235$$

4. Ans (B)

$$\frac{x-3}{1} = \frac{y-5}{-1} = -2 \frac{(3-5+1)}{1^2+1^2}$$

$$(x, y) = (4, 4)$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A, C, D)

$$(A) \log_a a - \log_a b + \log_a b - \log_a a = 0$$

$$(B) \log_{2024} 2024 = 1$$

$$(C) \log_2 (\log_{16} n) = \log_2 (\log_4 n)$$

$$\Rightarrow \log_2 (\log_{16} n) = \frac{1}{2} \log_2 (\log_4 n)$$

$$\Rightarrow 2 \log_2 \log_{16} n = \log_2 \log_4 n$$

$$\Rightarrow \log_2 \log_{16}^2 n = \log_2 \log_4 n$$

$$\Rightarrow \log_{16}^2 n = \log_4 n$$

$$\Rightarrow (\log_4 n)^2 = \log_4 n$$

$$\Rightarrow \left(\frac{1}{2} \log_4 n \right)^2 = \log_4 n$$

$$\Rightarrow x^2 = 4x$$

$$x=0, 4$$

$$\log_4 n = 0, 4 \Rightarrow n = 4^0, 4^4$$

$$n = 4^4 \text{ is acceptable}$$

$$n = 2^8 = 256$$

$$(D) \log_{2024} \log_{2023} \log_{2022} \log_{2021} \log_{2020} x$$

$$\log_{2023} \log_{2022} \log_{2021} \log_{2020} x > 0$$

$$\log_{2022} \log_{2021} \log_{2020} x > 1$$

$$\log_{2021} \log_{2020} x > 2022$$

$$\log_{2020} x > 2021^{2022}$$

$$x > 2020^{2021^{2022}}$$

6. Ans (C, D)

$$f'(x) + g(x) = \cos x (\cos x - \sin x) + \sin x (-\sin x - \cos x)$$

$$f'(x) + g(x) = \cos^2 x - \sin^2 x - \cos x \sin x - \sin x \cos x$$

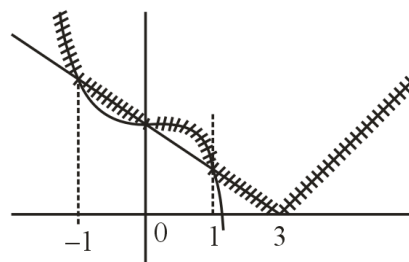
$$f'(x) + g(x) = \cos 2x - \sin 2x$$

$$\text{one possibility } f(x) = \frac{\sin 2x}{2}, g(x) = -\sin 2x$$

$$\text{another possibility } f(x) = -\sin^2 x, g(x) = \cos 2x$$

7. Ans (A, D)

Clearly $f(x)$ is non-derivable at $-1, 0, 1, 3$



PART-3 : MATHEMATICS

SECTION-I (iii)

8. **Ans (A)**

conceptual

apply tangent condition for given curves.

9. **Ans (A)**

(A) Solution is $xy^2 = 2y^5 + c$, (1, 1) lies on it

$$\Rightarrow c = -1 \Rightarrow xy^2 = 2y^5 - 1$$

$$y = 2 \Rightarrow x = \frac{63}{4} \Rightarrow [x] = 15$$

(B) Solution is $x^2y^2 + 1 = cy$, (1, 1) lies on it

$$\Rightarrow c = 2 \Rightarrow x^2y^2 + 1 = 2y$$

(C) Area = 7

(D) $f(x) = e^x$

$$\text{Area} = e - \frac{1}{e}$$

$$[\Delta] = 2$$

$$f(x) = e^{-x}$$

$$\text{Area} = e - \frac{1}{e}$$

$$[\Delta] = 2$$

$$f(x) = \begin{cases} e^{-x}, & (-\infty, 0) \\ e^x, & [0, \infty) \end{cases}$$

$$\text{Area} = (-e^{-x}) + (e^x)_0^1$$

$$= -1 + e^{-1} + e - 1$$

$$= 2e - 2$$

$$[\Delta] = 3$$

$$f(x) = \begin{cases} e^x, & (-\infty, 0) \\ e^{-x}, & [0, \infty) \end{cases}$$

$$\text{Area} = (e^x)_{-1}^0 + (-e^{-x})_0^1$$

$$= 1 - \frac{1}{e} + 1 - \frac{1}{e} = 2 - \frac{2}{e}$$

$$[\Delta] = 1$$

10. **Ans (A)**

(A) $y = 2^{-x} + 2^{-x^3}$ is decreasing function $y = x$ is

increasing function they intersect once.

$$(B) \frac{2.2 \cos\left(\frac{\theta_1 + \theta_2}{2}\right) \sin\left(\frac{\theta_1 - \theta_2}{2}\right)}{2 \sin\left(\frac{\theta_1 + \theta_2}{2}\right) \sin\left(\frac{\theta_1 - \theta_2}{2}\right)} \tan \theta$$

$$\Rightarrow 2 \cot\left(\frac{\theta_1 + \theta_2}{2}\right) \tan \theta$$

$$\Rightarrow 2 \cot \theta \cdot \tan \theta$$

$$\Rightarrow 2$$

$$(C) \sqrt{6 - 2\sqrt{5}} + \sqrt{6 + 2\sqrt{5}} = (\sqrt{5} - 1) + (\sqrt{5} + 1) = 2\sqrt{5}$$

$$= 6 \times \log_{20}(2\sqrt{5}) = 6 \times \frac{1}{2} = 3$$

$$(D) \log_{\frac{1}{e}} |x - 4| > -1 \Rightarrow \log_e |x - 4| < 1$$

$$|x - 4| < e \Rightarrow x \in (4 - e, 4 + e)$$

11. **Ans (B)**

(A) Using king's property $I = -I \Rightarrow I = 0$

$$(B) I_1 + I_1 = \int_0^{\pi/2} \left(\cos\left(\frac{2\pi}{3}\right) \sin^2 x + \cos\left(\frac{2\pi}{3}\right) \cos^2 x \right) dx$$

etc ...

$$2I_1 = I_2 \Rightarrow \frac{I_2}{I_1} = 2$$

(C)

$$\int (x+1)(x^2+2x)^{1/10} dx = \frac{1}{2} \frac{(x^2+2x)^{11/10}}{11/10} + c = \frac{5}{11} (x^2+2x)^{11/10}$$

$$A = 5, b = 10 \Rightarrow \frac{b}{a} = 2$$

(D)

$$\int x^2 (2x \cos 2x + 3 \sin 2x) dx = \int d(x^3 \sin 2x) = x^3 \sin 2x + c$$

$$\therefore c = 0 \quad \therefore f(x) = x^3 \Rightarrow x = 0 \text{ (or) } \sin 2x = 1$$

$$x = 0, x = \frac{\pi}{4}, \frac{5\pi}{4}$$

No of solution = 3

PART-3 : MATHEMATICS

SECTION-II

1. Ans (518)

Let the second term of each series be x . Then, the common ratio is $\frac{1}{8x}$ and the first term is $8x^2$.

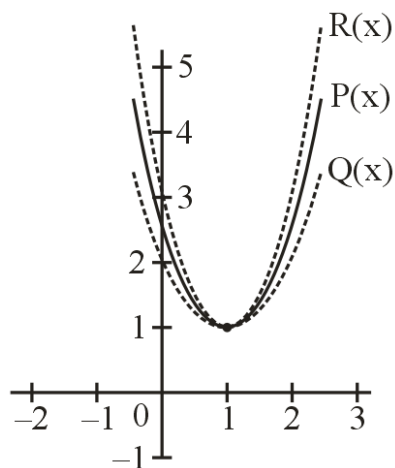
So, the sum is $\frac{8x^2}{1 - \frac{1}{8x}} = 1$. Thus, $64x^3 - 8x + 1$

$$= (4x - 1)(16x^2 + 4x - 1) = 0 \Rightarrow x = \frac{1}{4}, \frac{-1 \pm \sqrt{5}}{8}.$$

The only solution in the appropriate form is

$$x = \frac{\sqrt{5} - 1}{8}. \text{ Therefore, } 100m + 10n + p = 518.$$

2. Ans (406)



Let $Q(x) = x^2 - 2x + 2$, $R(x) = 2x^2 - 4x + 3$.

Completing the square,

We have $Q(x) = (x - 1)^2 + 1$ and

$$R(x) = 2(x - 1)^2 + 1,$$

so it follows that $P(x) \geq Q(x) \geq 1$ for all x (by the trivial inequality).

Also, $1 = Q(1) \leq P(1) \leq R(1) = 1$,

so $P(1) = 1$ and P obtains its minimum at the point $(1, 1)$.

Then $P(x)$ must be of the form $c(x - 1)^2 + 1$ for some constant c ,

substituting $P(11) = 181$ yields $c = \frac{9}{5}$

$$\text{Finally, } P(16) = \frac{9}{5} \cdot (16 - 1)^2 + 1 = 406.$$

3. Ans (1)

$$\frac{1}{\sqrt{xn^2 + n}} \leq \frac{1}{\sqrt{xn^2 + k}} \leq \frac{1}{\sqrt{xn^2}}, \sum_{k=1}^n \frac{1}{\sqrt{xn^2 + n}} \geq \sum_{k=1}^n \frac{1}{\sqrt{xn^2 + k}} \geq \sum_{k=1}^n \frac{1}{\sqrt{xn^2}}$$

$$\sum_{k=1}^n \frac{1}{\sqrt{xn^2}} = n \cdot \frac{1}{\sqrt{xn^2}} = \frac{1}{\sqrt{x}}$$

$$\sum_{k=1}^n \frac{1}{\sqrt{xn^2 + n}} = n \cdot \frac{1}{\sqrt{xn^2 + n}} = \frac{1}{\sqrt{x + \frac{1}{n}}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{x + \frac{1}{n}}} = \frac{1}{\sqrt{x}}$$

$$L = \lim_{x \rightarrow \infty} \sqrt{x} \sin \left(\sin \left(\sin \frac{1}{\sqrt{x}} \right) \right) =$$

$$= \lim_{x \rightarrow \infty} \frac{\sin \left(\sin \left(\sin \frac{1}{\sqrt{x}} \right) \right)}{\sin \left(\sin \frac{1}{\sqrt{x}} \right)} \cdot \frac{\sin \left(\sin \frac{1}{\sqrt{x}} \right)}{\sin \frac{1}{\sqrt{x}}} \cdot \frac{\sin \frac{1}{\sqrt{x}}}{\frac{1}{\sqrt{x}}} = 1.1.1 = 1$$

4. Ans (9)

$$A_n = \left[\frac{2i + j}{3^{2n}} \right] = \frac{1}{3^{2n}} (2i + j)$$

$$\Rightarrow 3^n A_n = \frac{1}{3^n} (2i + j)$$

$$\text{tr}(3^n A_n) = \text{tr} \left(\frac{1}{3^n} (2i + j) \right) = \frac{1}{3^n} (3 + 6 + 9) = \frac{18}{3^n}$$

$$\ell = \lim_{n \rightarrow \infty} \left(6 + 2 + \frac{2}{3} + \dots + \frac{18}{3^n} \right) = \frac{6}{1 - \frac{1}{3}} = 9$$

5. Ans (1)

$$\frac{T_4}{T_2} = \frac{5C_3 \left(\frac{1}{x} \right)^2 \cdot x^3 \cdot \tan^3 x}{5C_1 \left(\frac{1}{x} \right)^4 x \tan x} = \frac{2}{27} \pi^4;$$

$$2x^4 \tan^2 x = \frac{2\pi^4}{27};$$

$$x^4 \tan^2 x = \left(\frac{\pi}{3} \right)^4 (\sqrt{3})^2; x = \frac{\pi}{3}$$

6. Ans (35)

$$f(x) = x^\alpha + 3x^\beta$$

$$f'(x) = \alpha x^{\alpha-1} + 3\beta x^{\beta-1}$$

$$\Rightarrow f'(x) > 0 \quad \forall x \in (0, 1)$$

Hence $f(x)$ is increasing and concave upwards curve. So $\lambda = (0 + 1)/2 = 1/2$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE-SCORE

03-12-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	A	C	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C	A,B,C	A,C			
SECTION-II (i)	Q.	1	2	3	4		
	A.	10.00	108.00	8.00	2.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	4	8	37	25	5	13

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	C	D		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C,D	A,C	A,B,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	5.00	9.35	24.00	4.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	5	4	2	3	500	7

PART-3 : MATHEMATICS

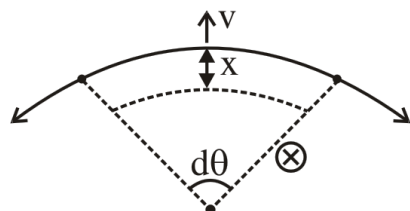
SECTION-I (i)	Q.	1	2	3	4		
	A.	A	C	B	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	C,D	A,B	B,C,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	-10.00	0.00	6.00	4.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	8	2	18	4	5	15

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)



$$\frac{2T d\theta}{2} = \frac{m}{2\pi a} \times a d\theta \times a_n$$

$$T = \frac{ma_n}{2\pi a}$$

$$T = \frac{Y A 2\pi x}{2\pi a}$$

$$-\frac{Y A x}{a} = \frac{ma_n}{2\pi}$$

$$a_n = \frac{-2\pi Y A}{ma} x$$

$$\omega_0 = \sqrt{\frac{2\pi Y A}{am}}$$

$$\phi = B\pi r^2$$

$$\frac{d\phi}{dt} = B2\pi r \frac{dr}{dt}$$

$$i = \frac{1}{R} \times 2\pi r B \frac{dr}{dt}$$

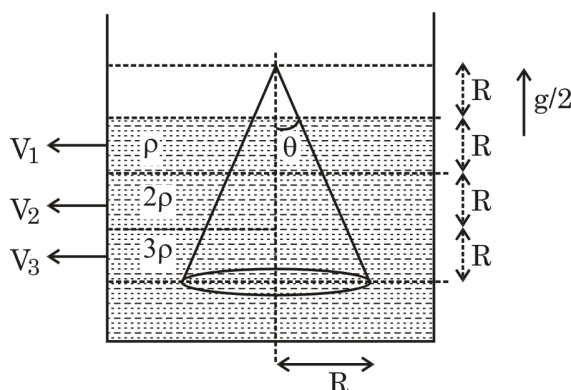
$$F = \frac{2\pi a B^2}{R} v a d\theta$$

$$F_{\text{net}} = -T d\theta - \frac{2\pi a^2 B^2}{R} v d\theta = \frac{m}{2\pi a} \times a d\theta a_n$$

$$\frac{d^2x}{dt^2} + \frac{4\pi^2 a^2 B^2}{mR} \frac{dx}{dt} + \frac{2\pi Y A x}{ma} = 0$$

$$\omega = \sqrt{\frac{2\pi Y A}{ma} - \frac{4\pi^4 a^4 B^4}{m^2 R^2}}$$

3. Ans (C)



$$\tan \theta = \frac{1}{4}$$

$$v_1 = \frac{\pi}{3} [x_2^2 \cdot 2R - x_1^2 R] = \frac{7\pi}{48} R^3$$

$$v_2 = \frac{\pi}{3} [x_3^2 \cdot 3R - x_2^2 \cdot 2R] = \frac{19\pi}{48} R^3$$

$$v_3 = \frac{\pi}{3} [R^2 \cdot 4R - x_3^2 \cdot 3R] = \frac{37\pi}{48} R^3$$

$$\text{So, } m = v_1 \rho + v_2 (2\rho) + v_3 (3\rho)$$

$$\rho_0 \frac{\pi R^2}{3} \times 4R \left(\frac{3g}{2} \right) = \rho \frac{\pi R^3}{48} (156) \left(\frac{3g}{2} \right)$$

$$\text{Density of cone, } \rho_0 = \frac{39}{16} \rho$$

PART-1 : PHYSICS

SECTION-I (ii)

6. **Ans (A,B,C)**

$$\frac{m_0 v^2}{r_n} = \frac{1}{4\pi\epsilon_0} \frac{ze^2}{r_n^2} \text{ (Where } m_0 = 50 \text{ m)} \dots\dots(i)$$

$$mv_0 r_n = \frac{n}{2} \frac{h}{\pi} \dots\dots(ii)$$

$$r_n = \frac{n^2 h^2 \epsilon_0}{200\pi m e^2}$$

$$E_n = \frac{-13.6Z^2}{n^2} \times 50 \text{ eV}$$

$$E_4 = -680 \text{ eV}$$

$$E_1 = -10880 \text{ eV}$$

$$\Delta E = E_4 - E_1 = 10,200 \text{ eV}$$

$$\lambda = \frac{1240}{\Delta E} = 1.21 \text{ \AA}$$

7. **Ans (A,C)**

For the first interference maximum

$$2d \sin \theta = \frac{\lambda}{2}$$

$$\Rightarrow \sin \theta = \frac{\lambda}{4d} = \frac{10^{-3}}{4}$$

$$\frac{y}{D} = \frac{10^{-3}}{4}$$

$$y = \frac{1}{4} \text{ mm}$$

$$\text{Fringe width, } \omega = \frac{\lambda D}{2d} = 0.5 \text{ mm}$$

$$2d \sin \theta = \frac{\lambda}{4}$$

$$\Rightarrow \sin \theta = \frac{\lambda}{8d}$$

$$\therefore y = \frac{\lambda D}{8d} = \frac{1}{8} \text{ mm}$$

PART-1 : PHYSICS

SECTION-II (i)

2. **Ans (108.00)**

Condition of no. slipping $\omega R - 6\omega_0 R = 4\omega_0 R$

$$\omega = 10 \omega_0$$

Acceleration of Point A

$$\vec{a}_A = \vec{a}_{A,C} + \vec{a}_C \quad \uparrow 12\omega_0^2 R$$

$$a_A = 88\omega_0^2 R \downarrow \quad \downarrow \overset{A}{(10\omega_0)^2 R}$$

Acceleration of point B with respect to surface

$$(\vec{a}_B - \vec{a}_{\text{surface}}),$$

$$\vec{a}_B = \vec{a}_{B,C} + \vec{a}_C$$

$$\vec{a}_S = \omega_0^2 4R = 4\omega_0^2 R$$

$$\vec{a}_{PS} = (112 - 4)\omega_0^2 R = 108\omega_0^2 R$$

PART-1 : PHYSICS

SECTION-II (ii)

9. **Ans (5)**

$$d\phi = (2\pi x dx) k x t^2$$

$$\phi = \frac{2}{3} k t^2 \pi x^3$$

$$\frac{d\phi}{dt} = \frac{4k\pi x^3}{3}$$

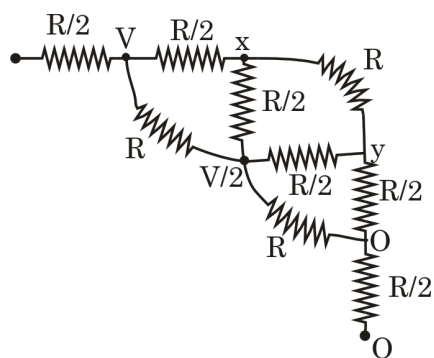
$$E 2\pi x = \frac{4k\pi x^3}{3}; E = \frac{2}{3} k t x^2;$$

$$d\tau = \left(\frac{2}{3} k t x^2 \right) \frac{Q}{\pi R^2} (2\pi x dx) x$$

$$\int d\tau = \frac{4}{3} \frac{k t Q}{R^2} \int_0^R x^4 dx \Rightarrow \tau = \frac{4}{3} \frac{k t Q}{R^2} \frac{R^5}{5}$$

At $t = 15 \text{ sec}$, $\tau = 5 \text{ N-m}$

10. Ans (13)



$$\frac{x-V}{\frac{R}{2}} + \frac{x-y}{R} + \frac{x-\frac{V}{2}}{\frac{R}{2}} = 0$$

$$\frac{y-x}{R} + \frac{y-\frac{V}{2}}{R/2} + \frac{y-0}{R/2} = 0$$

$$2(x-V) + x-y + 2(x-\frac{V}{2}) = 0$$

$$y-x + 2(y-\frac{V}{2}) + 2(y-0) = 0$$

$$5x - y - 3V = 0$$

$$5y - x - V = 0, \times 5$$

$$24y - 8V = 0, y = \frac{V}{3}$$

$$5x - \frac{V}{3} - 3V = 0$$

$$5x - \frac{10V}{3} = 0$$

$$x = \frac{2V}{3}$$

$$i = \frac{V - \frac{2V}{3}}{R/2} + \frac{V - \frac{V}{2}}{R} = \frac{\frac{V}{3}}{\frac{R}{2}} + \frac{V}{2R} = \frac{2V}{3R} + \frac{V}{2R}$$

$$i = \frac{7V}{6R} \Rightarrow V = \frac{6R}{7}i$$

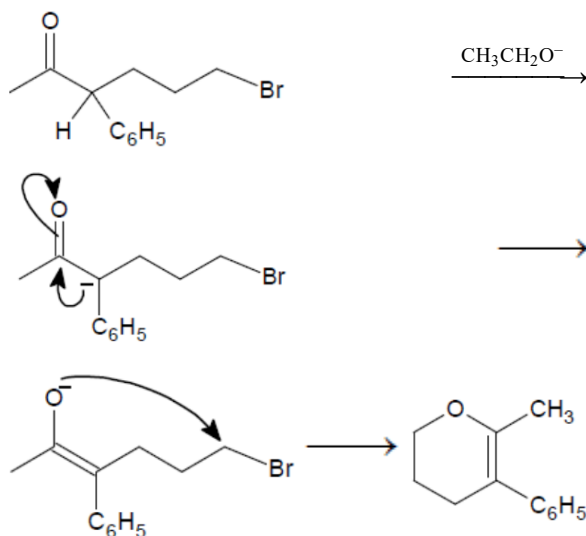
$$R_{eq} = \frac{6R}{7}$$

$$R' = \frac{13R}{7}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B)



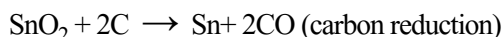
2. Ans (B)

$$\begin{aligned} \frac{x}{m} &= kP^{1/n} & \log k &= 0.3010 \\ &= 2 \times (1)^{\sqrt{3}} & k &= 2 \\ &= 2 & \frac{1}{n} &= \tan 60^\circ = \sqrt{3} \end{aligned}$$

4. Ans (D)

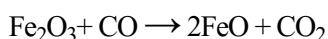
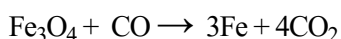
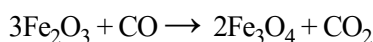
(A) Cassiterite contains iron and manganese tungstate, called as Wolframite. They have magnetic properties.

(B) Purified ore that contains 70% SnO_2 is called black tin.



(C) Roasting is done at moderate temperature in presence of air in reverberatory furnace. Then air supply is stopped and temperature is increased to melt the mass when self-reduction takes place.

(D) At 500 - 800K (lower temperature range in the upper part of blast furnace)



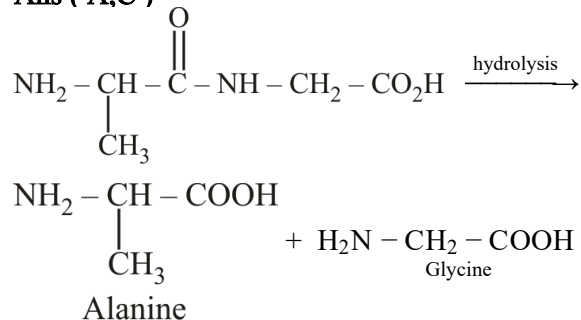
At 900 - 1500K (higher temperature range in the lower part of blast furnace:



PART-2 : CHEMISTRY

SECTION-I (ii)

6. Ans (A,C)



PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (5.00)

$$\text{pH} = \text{pK}_a + \log \frac{[\text{B}^-]}{[\text{HB}]}$$

$$5 = \text{pKa}$$

2. Ans (9.35)

$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

6 = pKa

$$C = \frac{0.1}{2}$$

At equivalence point :

$$[\text{H}^+] = \sqrt{\frac{K_w \cdot K_a}{C}}$$

$$[\text{H}^+] = \sqrt{\frac{10^{-14} \times 10^{-6}}{\frac{0.1}{2}}}$$

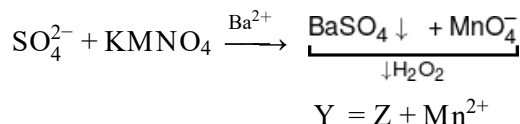
$$[\text{H}^+] = \sqrt{2 \times 10^{-19}}$$

$$\text{pH} = \frac{1}{2} (19 - \log 2)$$

$$\text{pH} = \frac{1}{2} \quad (18.7)$$

pH = 9.35

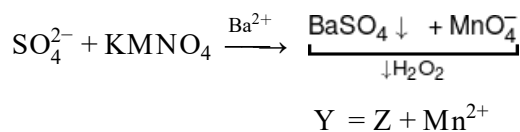
3. Ans (24.00)



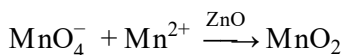
[Y : Pink due to adsorption of KMnO_4]

$$\text{SO}_4^2- : \text{total orbitals in bonding} = 14$$

4. Ans (4.00)



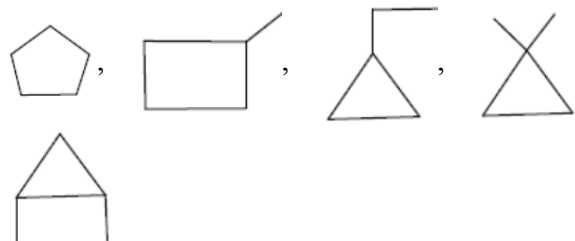
[Y : Pink due to adsorption of KMnO_4]

$$\text{SO}_4^2- : \text{total orbitals in bonding} = 14$$


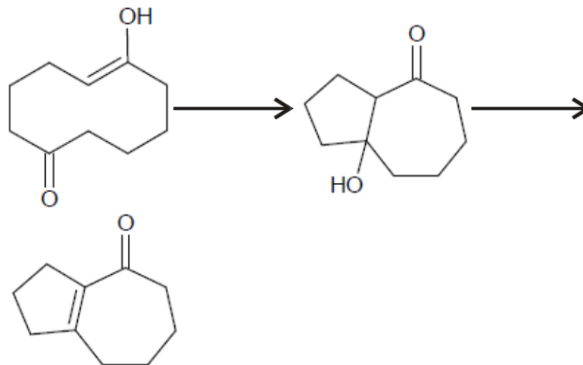
PART-2 : CHEMISTRY

SECTION-II (ii)

5. Ans (5)

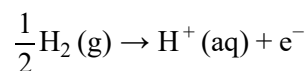


6. **Ans (4)**

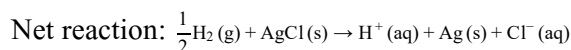
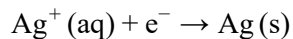
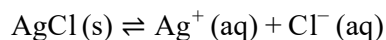


8. Ans (3)

At anode



at cathode



$$Q = \frac{[\text{H}^+][\text{Cl}^-]}{\sqrt{P_{\text{H}_2}}} = [\text{H}^+][\text{Cl}^-]$$

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.06}{1} \log Q$$

$$\Rightarrow 0.47 = 0.23 - 0.06 \log [\text{H}^+][\text{Cl}^-]$$

$$\frac{0.24}{0.06} = -\log \{0.1 \times [\text{H}^+]\}$$

$$4 = \log \frac{1}{0.1 \times [\text{H}^+]} \Rightarrow 10^4 = \frac{1}{0.1 \times [\text{H}^+]}$$

$$\Rightarrow [\text{H}^+] = 10^{-3}$$

$$\Rightarrow \text{pH} = 3$$

$$\Rightarrow 0.53 = 0.23 - 0.060 \log [\text{H}^+][\text{Cl}^-]$$

$$\frac{0.30}{0.06} = \log \frac{1}{0.1 \times [\text{H}^+]}$$

$$\Rightarrow 5 = \log \frac{1}{0.1 \times [\text{H}^+]} \Rightarrow 10^5 = \frac{1}{0.1 \times [\text{H}^+]}$$

$$\therefore [\text{H}^+] = 10^{-4}$$

$$\therefore \text{pH} = 4$$

If there is 'x' mole of acid and 'y' mole of NaOH in its 20mL.

then,

$$3 = \text{p}K_a + \log \frac{y}{x-y} \quad \dots\dots\dots (i) \quad \text{and}$$

$$4 = \text{p}K_a + \log \frac{2.5y}{x-2.5y} \quad \dots\dots\dots (ii)$$

$$\Rightarrow I = \log \frac{2.5(x-y)}{x-2.5y}$$

$$\Rightarrow 10 = \frac{2.5x-2.5y}{x-2.5y} \Rightarrow 10x-25y = 2.5x-2.5y$$

$$x = \frac{22.5y}{7.5} = 3y$$

From equation (i)

$$3 = \text{p}K_a + \log \frac{y}{x-y}$$

$$3 = \text{p}K_a + \log \frac{y}{2y}$$

$$\therefore \text{p}K_a = 3.3$$

9. Ans (500)

$$\Delta H_{\text{reaction}}^{\circ} = -375 - (-290)$$

$$= -85 \text{ kJ/mole}$$

$$\Delta S^{\circ} = 360 - 220 - 310$$

$$= -170 \text{ J/mole/K}$$

For equilibrium $\Delta G^{\circ} = 0$

$$\Delta H^{\circ} = T \Delta S^{\circ}$$

$$T = \frac{\Delta H^{\circ}}{\Delta S^{\circ}}$$

$$= \frac{-85}{-170} \times 1000 = 500 \text{ K}$$

10. Ans (7)

$$\text{For } A \rightarrow B \quad 600 V_1^{\gamma-1} = 60 V_2^{\gamma-1} \quad (\gamma = 5/3)$$

(Reversible adiabatic)

$$\Rightarrow 600 (V_1)^{2/3} = 60 (V_2)^{2/3}$$

$$\Rightarrow 10 = \left(\frac{V_2}{V_1} \right)^{2/3}$$

$$\Rightarrow 10 = \left(\frac{V_2}{10} \right)^{2/3}$$

$$\Rightarrow V_2 = 10(10)^{3/2} = 10^{5/2}$$

$$\text{Now, } q_{\text{net}} = RT_2 \ln 10 = 60 R \ln 10 = q_{AB} + q_{BC}$$

$$\therefore q_{AB} = 0$$

$$\Rightarrow q_{BC} = 60 R \ln 10 = 60 R \ln \frac{V_3}{V_2}$$

[$\therefore B \rightarrow C$ is reversible isothermal]

$$\Rightarrow 60 R \ln 10 = 60 R \ln \left(\frac{V_3}{10^{5/2}} \right)$$

$$\Rightarrow \log 10 = \log V_3 - \frac{5}{2}$$

$$\Rightarrow \log V_3 = \frac{7}{2}$$

$$\Rightarrow 2 \log V_3 = 7$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)

$$x^3 - 6x^2 + 3px - 2p = 0 \begin{cases} x_1 \\ x_2 \\ x_3 \end{cases}$$

$$A.M. = \frac{x_1 + x_2 + x_3}{3} = \frac{6}{3} = 2$$

$$H.M. = \frac{3x_1x_2x_3}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3}} = \frac{3x_1x_2x_3}{\sum x_1x_2} = 2$$

$$\therefore A.M = H.M. \Rightarrow x_1 = x_2 = x_3 =$$

$$\sin^{-1}\left(\frac{1}{x_1} + \frac{1}{x_2}\right) + \cos^{-1}\left(\frac{1}{x_2} + \frac{1}{x_3}\right) - \tan^{-1}\left(\frac{1}{x_3} + \frac{1}{x_1}\right) \\ = \frac{\pi}{2} + 0 - \frac{\pi}{4} = \frac{\pi}{4}$$

2. Ans (C)

$$2^r \cdot \frac{20!}{r!(20-r)!} \cdot \frac{(20-r)!}{10!(10-r)!} = \frac{20!}{10!10!} \times \frac{10!}{r!(10-r)!} \times 2^r$$

3. Ans (B)

$$\cos(x/y) \left(\frac{ydx - xdy}{y^2} \right) = xy d(xy)$$

$$\cos(x/y) d(x/y) = xy d(xy)$$

Integrate both sides.

4. Ans (C)

$$\lim_{x \rightarrow 1} \left(\frac{k}{\ln x} - \frac{k}{x-1} \right) = 2$$

Put $x = 1 + h$

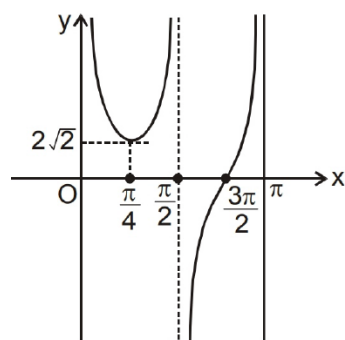
$$\Rightarrow \lim_{h \rightarrow 0} \frac{k(h - \ln(1+h))}{h \cdot \ln(1+h)} = 2$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{k \left[h - \left(h - \frac{h^2}{2} + \dots \right) \right]}{h^2} = 2 \Rightarrow \frac{k}{2} = 2 \Rightarrow k = 4$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (C,D)



Graph of $y = \sec x + \csc x$

6. Ans (A,B)

Conceptual

7. Ans (B,C,D)

Parabola touches x-axis at (2,0) and y-axis at (0,3)

tangent at vertex is line joining (2,0) and (0,3) which

is $3x + 2y = 6$

(0,0) lies on directrix. Hence directrix is $3x + 2y = 0$.

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (-10.00)

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} - \sqrt{x^4 + 2x^3 - bx^2 + 3x + c} \right) = 4$$

$$\lim_{x \rightarrow \infty} \left(\frac{(x^4 + ax^3 + 3x^2 + bx + 2) - (x^4 + 2x^3 - bx^2 + 3x + c)}{\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} + \sqrt{x^4 + 2x^3 - bx^2 + 3x + c}} \right) = 4$$

$$\lim_{x \rightarrow \infty} \frac{(a-2)x^3 + (3+b)x^2 + (b-3)x + (2-c)}{x^2 \left(\sqrt{1 + \frac{a}{x} + \frac{3}{x^2} + \frac{b}{x^3} + \frac{2}{x^4}} + \sqrt{1 + \frac{2}{x} - \frac{b}{x^2} + \frac{3}{x^3} + \frac{c}{x^4}} \right)} = 4$$

$$\Rightarrow a = 2, \frac{3+b}{2} = 4 \Rightarrow b = 5 \text{ and } c \in \mathbb{R}.$$

2. Ans (0.00)

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} - \sqrt{x^4 + 2x^3 - bx^2 + 3x + c} \right) = 4$$

$$\lim_{x \rightarrow \infty} \left(\frac{(x^4 + ax^3 + 3x^2 + bx + 2) - (x^4 + 2x^3 - bx^2 + 3x + c)}{\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} + \sqrt{x^4 + 2x^3 - bx^2 + 3x + c}} \right) = 4$$

$$\lim_{x \rightarrow \infty} \frac{(a-2)x^3 + (3+b)x^2 + (b-3)x + (2-c)}{x^2 \left(\sqrt{1 + \frac{a}{x} + \frac{3}{x^2} + \frac{b}{x^3} + \frac{2}{x^4}} + \sqrt{1 + \frac{2}{x} - \frac{b}{x^2} + \frac{3}{x^3} + \frac{c}{x^4}} \right)} = 4$$

$$f(x) - g(x) > 1 \quad \forall x \in \mathbb{R}$$

$$(x^4 + 2x^3 + 3x^2 + 5x + 2) - (x^4 + 2x^3 - 5x^2 + 3x + c) > 1$$

$$\Rightarrow 8x^2 + 2x + 2 - c > 1$$

$$\Rightarrow 8x^2 + 2x + 1 - c > 0 \quad \forall x \in \mathbb{R}$$

3. Ans (6.00)

Let $N = x_1x_2x_3\dots x_{10}$ be one of such number
 $\because \sum x_i = 45$, hence 'N' is divisible by 9, but 11111
 is

not divisible by 9. Hence N should be divisible
 $11111 \times 9 = 99999$

$$\begin{aligned} \text{Now } x_1x_2x_3\dots x_{10} &= x_1x_2x_3x_4x_5 \times 10^5 + \\ & x_6x_7x_8x_9x_{10} \\ &= x_1x_2x_3x_4x_5 \cdot 99999 + (x_1x_2x_3x_4x_5) + \\ & (x_6x_7x_8x_9x_{10}) \end{aligned}$$

$$\begin{aligned} \text{Now } x_1x_2x_3x_4x_5 + x_6x_7x_8x_9x_{10} &< (2 \times 99999) \\ \Rightarrow (x_1x_2x_3x_4x_5 + x_6x_7x_8x_9x_{10}) &= 99999 \\ \Rightarrow x_1 + x_6 = x_2 + x_7 = x_3 + x_8 = x_4 + x_9 \\ &= x_5 + x_{10} = 9 \end{aligned}$$

Smallest number = 1023489765

4. Ans (4.00)

Let $N = x_1x_2x_3\dots x_{10}$ be one of such number
 $\because \sum x_i = 45$, hence 'N' is divisible by 9, but 11111
 is

not divisible by 9. Hence N should be divisible
 $11111 \times 9 = 99999$

$$\begin{aligned} \text{Now } x_1x_2x_3\dots x_{10} &= x_1x_2x_3x_4x_5 \times 10^5 + \\ & x_6x_7x_8x_9x_{10} \\ &= x_1x_2x_3x_4x_5 \cdot 99999 + (x_1x_2x_3x_4x_5) + (x_6x_7x_8x_9x_{10}) \end{aligned}$$

$$\begin{aligned} \text{Now } x_1x_2x_3x_4x_5 + x_6x_7x_8x_9x_{10} &< (2 \times 99999) \\ \Rightarrow (x_1x_2x_3x_4x_5 + x_6x_7x_8x_9x_{10}) &= 99999 \\ \Rightarrow x_1 + x_6 = x_2 + x_7 = x_3 + x_8 = x_4 + x_9 \\ &= x_5 + x_{10} = 9 \end{aligned}$$

Biggest number = 9876501234

PART-3 : MATHEMATICS

SECTION-II (ii)

5. Ans (8)

$$\because \Delta = [(a_0^3 + a_1^3 + a_2^3) - 3a_0a_1a_2]$$

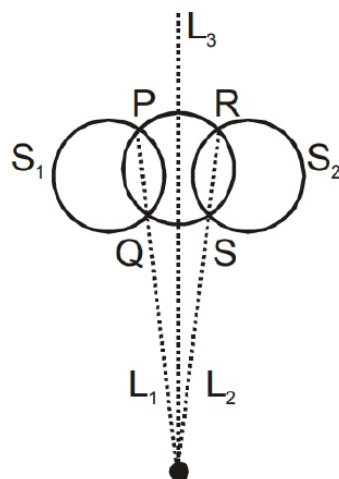
$D = 0$ and $a_0 = a_1 + a_2 \neq 0 \Rightarrow a_0 = a_1 = a_2$
 putting $x = 0$

$$a_0 = 1 \Rightarrow a_1 = a_2 = 1$$

coefficient of $x = 4a$ and coefficient of
 $x^2 = 4b + 6a^2$

$$1 = 4a = 6a^2 + 4b \Rightarrow a = \frac{1}{4}; b = \frac{5}{32}$$

6. Ans (2)



$$L_1 : x + 2y + c = 0 \quad \dots(1)$$

$$L_2 : 12x - 6y = 41 \quad \dots(2)$$

$$S_1 : x^2 + y^2 - 4 = 0$$

$$S_2 : x^2 + y^2 - 4x - 2y + 1 = 0$$

$L_1 \Rightarrow$ Radical axes of S_1 and circle passing P, Q, R, S

$L_2 \Rightarrow$ Radical axes of S_2 and circle passing P, Q, R, S

$L_3 \Rightarrow$ Radical axes of S_1 and S_2 given by $S_1 - S_2 = 0$

$$\Rightarrow 4x + 2y - 5 = 0 \quad \dots(3)$$

As radical axes of 3 circles are concurrent, so

$$\Rightarrow \begin{vmatrix} 1 & 2 & c \\ 12 & -6 & -41 \\ 4 & 2 & -5 \end{vmatrix} = 0 \Rightarrow c = 2$$

7. Ans (18)

$$\begin{aligned} \text{Let } T_r &= \left(\frac{2r+5}{r(r+1)} \right) \left(\frac{3}{5} \right)^{r+1} = \left(\frac{5(r+1)-3r}{r(r+1)} \right) \left(\frac{3}{5} \right)^{r+1} \\ &= 3 \left[\frac{1}{r} \left(\frac{3}{5} \right)^r - \frac{1}{(r+1)} \left(\frac{3}{5} \right)^{r+1} \right] = 3(v_r - v_{r+1}) \end{aligned}$$

8. Ans (4)

$$x^4 - 7x^2 - 4x + 20 = (x^2 - 4)^2 + (x - 2)^2$$

$$x^4 + 9x^2 + 16 = (x^2 + 4)^2 + x^2$$

Take the curve $y = x^2$. Both square roots can be interpreted as distances and $|PA - PB|$ has maximum value AB.

9. Ans (5)

Both graphs are mirror reflections w.r.t $x = y$.

So, minimum distance is along the line with

slope '-1'

Now, $x^2 - 8x - 4y +$

$40 = 0$

Point of contact of its

tangent with slope '1'

$$\Rightarrow 2x - 8 - 4 \left(\frac{dy}{dx} \right) = 0 \left[\frac{dy}{dx} = 1 \right]$$

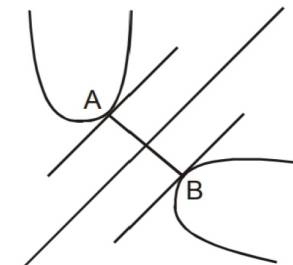
$$\Rightarrow x = 6 \Rightarrow y = 7 \Rightarrow A(x, y) \text{ is } (6, 7)$$

Similarly the point of contact of the tangent with

slope '1' to curve $y^2 - 8y - 4x + 40 = 0$ is

B(7,6)

$$AB = \sqrt{2}$$



10. Ans (15)

$$\text{Put } x = 1, 6 \int_1^1 g(t) dt = 3g(1) - 6 \Rightarrow g(1) = 2$$

Now, differentiate both the sides with respect to x ,

$$\therefore g(x) = xg'(x) - x^2 \quad [\text{b Leibnitz Rule}]$$

$$\text{or } \frac{xg'(x) - g(x)}{x^2} = 1 \Rightarrow \frac{d}{dx} \left(\frac{g(x)}{x} \right) = 1$$

$$\therefore \frac{g(x)}{x} = x + c \Rightarrow g(x) = x^2 + cx$$

$$\therefore g(1) = 2 \Rightarrow c = 1$$

$$\therefore g(x) = x^2 + x$$